OVERVIEW

Three experiments, one task

Comparison of the previous Graph-NTF run, the current paper-method adaptations, and the earlier log-mel SVM + EEGPT blend. All scores below are our experiments, not published paper accuracies.

> **Main finding:** CP6 + Naive Bayes reached **60.3% accuracy** on the matched 174-person evaluation. The SVM + EEGPT blend reached **70.4%** on a separate 27-person fixed test split; that score is exploratory and cannot be ranked directly against cross-validation.

Dataset and prediction target

The merged cohort contains **174 people: 71 AD, 42 FTD and 61 healthy controls**, from OpenNeuro ds004504 and usable BrainLat EEG. Every result predicts one of three classes for a person. Windows are repeated measurements, not additional independent patients.

Experiment	Evaluation	Accuracy	Balanced accuracy	FTD recall
A - Graph-NTF + SVM	174 people; 3 repeats x 5 outer folds; 3 inner folds	53.4%	47.2%	0.8%
B - CP6 + Naive Bayes (best adaptation)	174 people; 1 x 5 shared outer folds; 3 inner folds	60.3%	56.5%	31.0%
C - Log-mel SVM + EEGPT	121 train / 26 validation / 27 test; fixed split	70.4%	67.6%	50.0%

A reports the mean across three repeats. Page 2 uses only A's first repeat to match B exactly. C evaluates 27 test participants, including only six with FTD. Balanced accuracy is the average of AD, FTD and HC recall. FTD recall is the proportion of true FTD cases correctly identified.

What changed between the experiments?

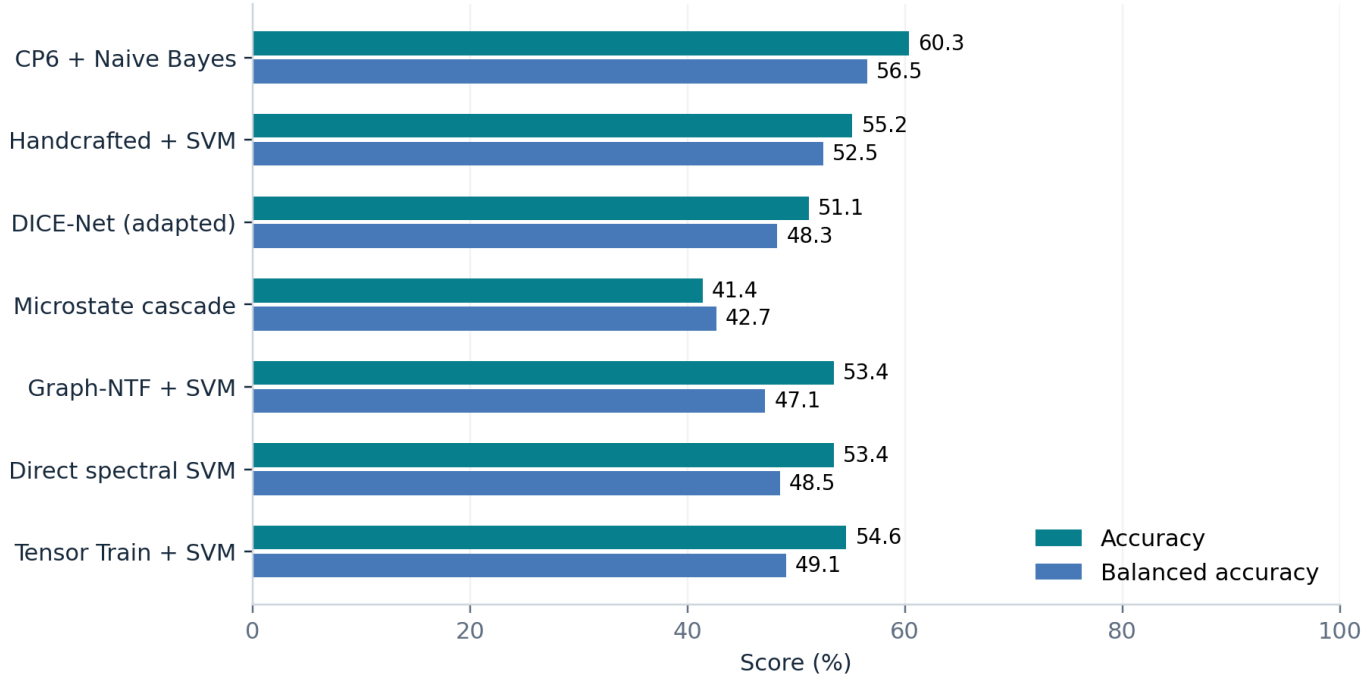
Run	Method
A	Nonnegative electrode-frequency-statistic tensor; graph-smoothed CP factors; site centering; SVM; probability calibration.
B	Four literature adaptations: CP6 + Naive Bayes, handcrafted SVM cascade, DICE-Net and microstate CNN cascade. Each evaluated on held-out people.
C	Site-normalized log-mel SVM combined with a partially fine-tuned EEGPT participant model using validation-selected probability blending.

Prepared on 03 November 2026. A and B are identified by their saved experiment outputs rather than an assumed wall-clock run date. No independent external cohort has been evaluated.

MATCHED CROSS-VALIDATION

Graph-NTF versus paper methods

Same 174 people, the same five outer test memberships, and exactly one held-out prediction per person. Previous controls use repeat 0 only. Representations and training procedures still differ.



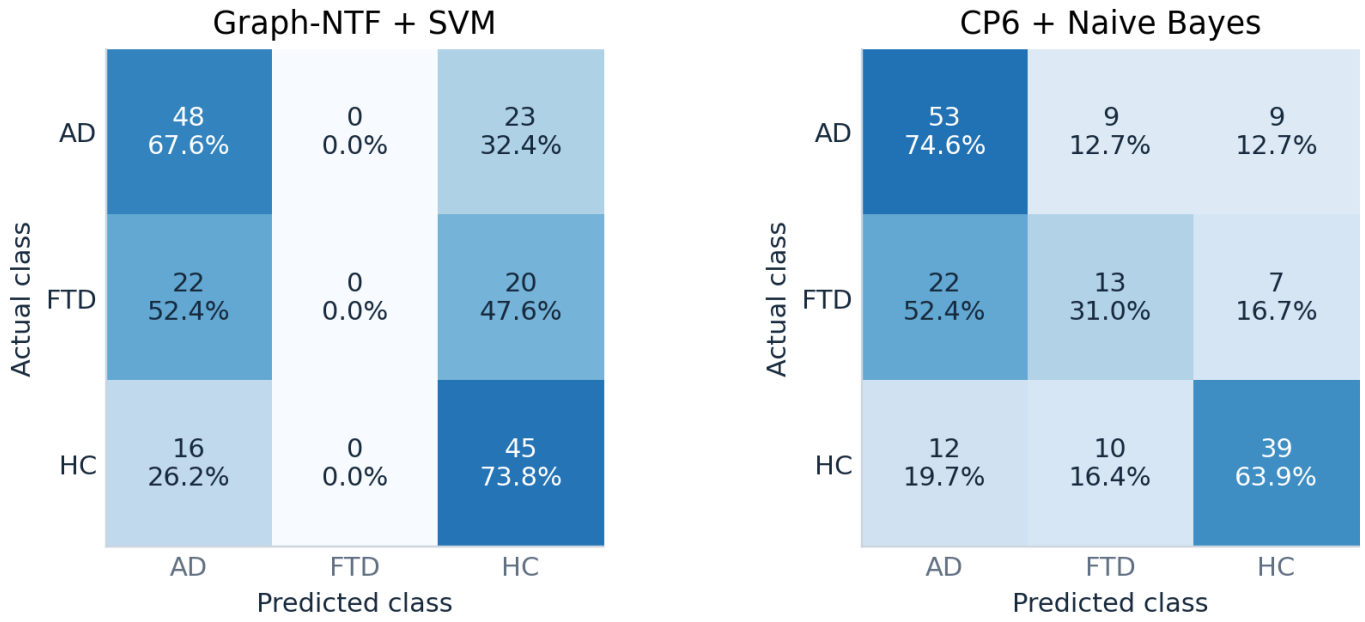
Method	Accuracy	Balanced acc.	Macro F1	FTD recall
CP6 + Naive Bayes	60.3%	56.5%	56.5%	31.0%
Handcrafted + SVM	55.2%	52.5%	51.8%	28.6%
DICE-Net adaptation	51.1%	48.3%	47.9%	26.2%
Microstate cascade	41.4%	42.7%	40.4%	45.2%
Graph-NTF + SVM	53.4%	47.1%	40.5%	0.0%
Direct spectral SVM	53.4%	48.5%	46.3%	11.9%
Tensor Train + SVM	54.6%	49.1%	46.3%	9.5%

CP6 gained 12 correct predictions: 105/174 versus 93/174 for Graph-NTF on these folds (+6.9 percentage points). The microstate cascade has higher FTD recall (45.2%) but lower overall accuracy (41.4%); catching more FTD cases alone does not imply a better overall classifier.

FTD AND DECISION BEHAVIOUR

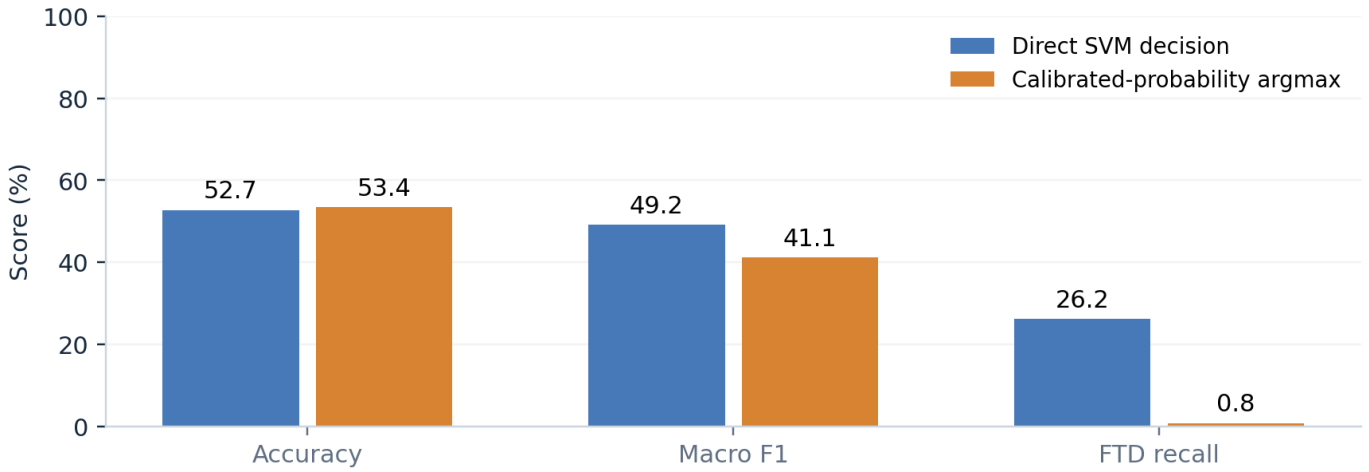
Why accuracy alone is insufficient

The top panels compare the same 174-person outer predictions. Cells show counts and percentages within the actual class; diagonal cells are correct predictions.



Matched five-fold FTD result: Graph-NTF classified 0/42 FTD participants correctly; CP6 classified 13/42. CP6 also improved AD from 48/71 to 53/71, while HC decreased from 45/61 to 39/61.

Calibration diagnostic: same saved Graph-NTF models



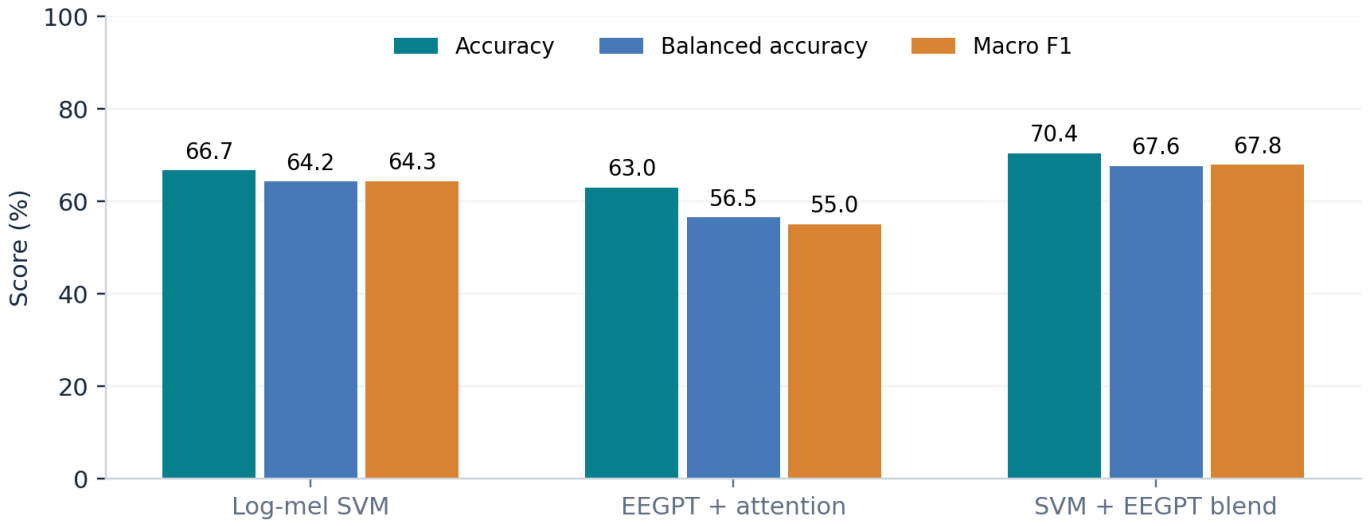
The lower chart averages all three Graph-NTF repeats. Direct SVM decisions gave **26.2% FTD recall**; calibrated-probability argmax gave **0.8%**. Accuracy changed from 52.7% to 53.4%, while macro F1 fell from 49.2% to 41.1%. This is a diagnostic replay, not a newly tuned improvement.

> **Decision-stage mismatch:** the original search selected SVMs using direct class predictions, but final reporting used calibrated-probability argmax. Future tuning should select the complete final decision rule inside training folds and track FTD precision, recall and errors on AD/HC.

SEPARATE FIXED-SPLIT BENCHMARK

Log-mel SVM + EEGPT blend

Exploratory result on 27 held-out participants: 11 AD, 6 FTD and 10 HC. The full cohort split was 121 training / 26 validation / 27 test participants.



Model	Correct	Accuracy	Macro F1	FTD recall
Log-mel SVM	18/27	66.7%	64.3%	50.0%
EEGPT + attention	17/27	63.0%	55.0%	16.7%
SVM + EEGPT blend	19/27	70.4%	67.8%	50.0%

How the two models were combined

The spectral branch used moderate-cleaning log-mel features, training-fitted site normalization, 20 PCA components and a linear SVM (C = 0.1). The EEGPT branch used pretrained EEG representations, participant window attention and partial fine-tuning.

Validation selected **50% SVM + 50% EEGPT** probability fusion after temperature scaling (SVM temperature 0.5; EEGPT temperature 0.75). The selected blend reached 73.1% validation accuracy and 71.0% validation macro F1.

> **What improved:** the blend corrected one additional test participant versus SVM alone. HC recall increased from 70% (7/10) to 80% (8/10). AD stayed at 8/11 and FTD stayed at 3/6. The gain was in healthy-control recognition, not FTD recognition.

Comparison limit: this 70.4% result comes from one small test split inspected during multiple development iterations. It is not an estimate from nested cross-validation and does not establish that the blend outperforms CP6 or Graph-NTF on unseen cohorts.

CONCLUSIONS AND EVIDENCE

What we can claim now

The report preserves the difference between observed scores, interpretable representations and evidence of generalization.

Interpretability

Pipeline	What is inspectable
Graph-NTF + SVM	Electrode, frequency and statistic profiles, plus participant activations. Graph regularization encourages spatial smoothness; it does not prove anatomical causality.
CP6 + Naive Bayes	Six electrode-frequency components and participant strengths. Component plots, stability and class contributions still need a dedicated interpretation analysis.
SVM + EEGPT blend	Defined spectral inputs plus a learned transformer representation. The blend has no shared small set of CP components explaining both branches.

Limits that materially affect the comparison

Optimization: 7/15 Graph-NTF outer fits and 3/5 CP6 outer fits had no optimizer-converged initialization within their frozen budgets. Convergence was not assumed.

Adaptations: the paper benchmark changes source-cohort details and some feature/architecture choices. DICE has a three-class head and corrected batch semantics; the SVM and microstate arms are documented reimplementations, not exact paper replications.

Data: both recent benchmarks inherit moderate preprocessing without explicit EEGLAB boundary-window rejection. This signal-quality issue is separate from participant train/test leakage.

Validation: the 174-person cohort has been explored repeatedly. The fixed-split blend is especially exploratory. No method has independent external validation.

Verification: the paper benchmark checked input/code hashes and disjoint participant memberships; all 20 saved fold models reproduced their held-out decisions.

Next controlled experiment

Prioritize FTD recall together with macro F1 and false positives. Select the complete decision/calibration rule inside training folds; improve tensor convergence; then vary representation, graph strength and classifier one at a time on shared participant splits.

Source artifacts (relative to C:\Projects\AIBSI)

1. neural_comparison/graph_ntf_svm/outputs/full/summary.json
2. neural_comparison/graph_ntf_svm/outputs/full/oof_predictions.json
3. neural_comparison/paper_benchmark/outputs/full_174/summary.json
4. neural_comparison/paper_benchmark/outputs/full_174/matched_previous_controls.json
5. neural_comparison/paper_benchmark/outputs/full_174/verification.json
6. neural_comparison/advanced_clean_spectral/outputs/deep_models/eegpt_attention/svm_blend_metrics.json
7. neural_comparison/advanced_clean_spectral/outputs/results/site_normalized_metrics.json
8. neural_comparison/advanced_clean_spectral/outputs/deep_models/eegpt_attention/metrics.json

Full method departures: neural_comparison/paper_benchmark/README.md. Paper references: Li et al. (2024), doi:10.1093/braincomms/fcae227; Rostamikia et al. (2024), doi:10.1007/s11571-024-10152-7; DICE-Net (2023), doi:10.1109/ACCESS.2023.3294618; Jiang et al. (2026), doi:10.3390/bios16050258.